

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (ISE) (Credit System) Degree Examinations
 May - June 2023

20ISE93 - OPERATIONS RESEARCH

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I

- | | Marks | BT* | CO* | PO* | | | | | | | | | | | | | | |
|---|-------|------|------|-------|-------|-----|-----|----|----|----|----|----|----|----|---|----|---|---|
| 1. a) List out the applications of Operations Research. | 6 | L*2 | 1 | 1 | | | | | | | | | | | | | | |
| b) Solve the following LP problem graphically:
Maximize $Z = -x_1 + 2x_2$ subject to the constraints
$x_1 - x_2 \leq -1$; $-0.5x_1 + x_2 \leq 2$ $x_1, x_2 \geq 0$. | 8 | L3 | 2 | 1 | | | | | | | | | | | | | | |
| c) Evening shift resident doctors in a government hospital work five consecutive days and have two consecutive days off. Their five days of work can start on any day of the week and their schedule rotates indefinitely. The hospital requires the following minimum number of doctors to work on the given days:
<table style="margin-left: 20px; border-collapse: collapse;"> <tr> <td>Sun</td><td>Mon</td><td>Tues</td><td>Wed</td><td>Thurs</td><td>Fri</td><td>Sat</td> </tr> <tr> <td>35</td><td>55</td><td>60</td><td>50</td><td>60</td><td>50</td><td>45</td> </tr> </table> No more than 40 doctors can start their five working days on the same day. Formulate this problem as an LP model to minimize the number of doctors employed by the hospital | Sun | Mon | Tues | Wed | Thurs | Fri | Sat | 35 | 55 | 60 | 50 | 60 | 50 | 45 | 6 | L3 | 2 | 1 |
| Sun | Mon | Tues | Wed | Thurs | Fri | Sat | | | | | | | | | | | | |
| 35 | 55 | 60 | 50 | 60 | 50 | 45 | | | | | | | | | | | | |
| 2. a) Use penalty (Big-M) method to solve the following LP problem.
Minimize $Z = 5x_1 + 3x_2$
subject to the constraints
$2x_1 + 2x_2 = 10$, $5x_1 + 2x_2 \geq 10$ $2x_1 + 4x_2 \leq 12$ and $x_1, x_2 \geq 0$. | 12 | L3 | 2 | 1 | | | | | | | | | | | | | | |
| b) Explain phases of Operations Research. | 8 | L2 | 2 | 1 | | | | | | | | | | | | | | |
| 3. a) Solve the following LPP using the Dual simplex method.
Minimize $Z = 2x_1 + x_2$ Subject to
$3x_1 + x_2 \geq 3$ $4x_1 + 3x_2 \geq 6$ $x_1 + 2x_2 \geq 3$, and $x_1 \geq 0, x_2 \geq 0$ | 12 | L3 | 2 | 1 | | | | | | | | | | | | | | |
| b) Show that dual of dual problem is primal problem, | 8 | L3 | 2 | 1 | | | | | | | | | | | | | | |

Unit – II

- | | | | | |
|---|---|----|---|---|
| 4. a) Explain the Transportation algorithm | 8 | L2 | 3 | 2 |
| b) Solve the following assignment problem using the Hungarian method.
A computer center has three expert programmers. The center wants three application programs to be developed. The head of the computer center, after carefully studying the programs to be developed, estimates the computer time in minutes required by the experts for the application programs as follows. | | | | |

		Programmers		
		A	B	C
Programmes	1	120	100	80
	2	80	90	110
	3	110	140	120

12 L3 4 2

5. a) The following table provides all the necessary information on the availability of supply to each warehouse, the requirement of each market, and the unit transportation cost (in Rs) from each warehouse to each market.

		Market				
		<i>P</i>	<i>Q</i>	<i>R</i>	<i>S</i>	<i>Supply</i>
Warehouse	<i>A</i>	6	3	5	4	22
	<i>B</i>	5	9	2	7	15
	<i>C</i>	5	7	8	6	8
	<i>Demand</i>	7	12	17	9	45

The shipping clerk of the shipping agency has worked out the following schedule, based on his own experience: 12 units from A to Q, 1 unit from A to R, 9 units from A to S, 15 units from B to R, 7 units from C to P and 1 unit from C to R.

(i) Check and see if the clerk has the optimal schedule.

(ii) Find the optimal schedule and minimum total transport cost.

15 L3 3 2
5 L2 4 2

- b) Explain Balanced and Unbalanced assignment problems.
6. a) A traveling salesman has to visit five cities. He wishes to start from a particular city, visit each city once and then return to his starting point. The traveling cost (in '000 Rs) of each city from a particular city is given below:

		To City				
		A	B	C	D	E
From City	A	∞	2	5	7	1
	B	6	∞	3	8	2
	C	8	7	∞	4	7
	D	12	4	6	∞	5
	E	1	3	2	8	∞

What should be the sequence of visit of the salesman so that the cost is minimum?

10 L3 4 2
10 L3 5 2

- b) Explain Hungarian method to solve assignment problem.

Unit – III

7. a) Explain PERT/ CPM network components.
- b) Draw the activity network and determine the total float and free float for each activity of the project.

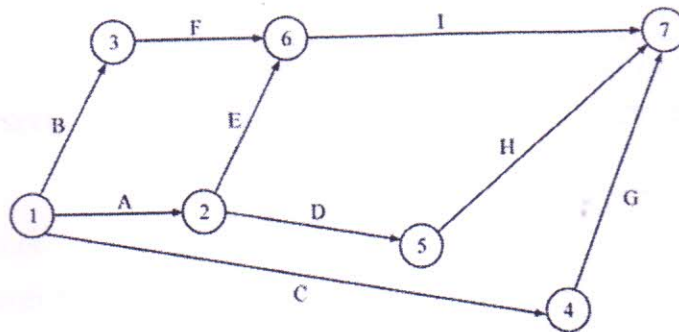
8 L2 5 2

Activity	Preceding Activity	Normal time (days)
1-2	-	20
1-3	-	25
2-3	1-2	10
2-4	1-2	12
3-4	1-3, 2-3	5
4-5	2-4, 3-4	10

12 L3 5 2
6 L2 5 2

8. a) Define Optimistic time, Pessimistic time and Most likely time.
- b) The following network diagram represents activities associated with a project:

Activities	A	B	C	D	E	F	G	H	I
Optimistic time, t_o	5	18	26	16	15	6	7	7	3
Pessimistic time, t_p	10	22	40	20	25	12	12	9	5
Most likely time, t_m	8	20	33	18	20	9	10	8	4



Determine the following:

- (i) Expected completion time and variance of each activity.
- (ii) The earliest and latest expected completion times of each event.
- (iii) The critical path.
- (iv) The probability of expected completion time of the project if the original scheduled time of completing the project is 41.5 weeks.
- (v) The duration of the project that will have 95 per cent chance of being completed.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
